



Name: Cohort/Batch: Date: Score:

SPRING PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Backend Technologies

TOTAL: 10 QUESTIONS

Q1 What is Spring and what problem in Backend Technologies does it solve?

Q6 How does Spring handle reliability and high availability?

Q2 What are the core components or concepts in Spring?

Q7 What observability does Spring provide out of the box?

Q3 How does Spring compare to its main alternatives?

Q8 What are common performance bottlenecks in Spring?

Q4 How do you get started with Spring for a new project?

Q9 What security best practices apply when running Spring?

Q5 What configuration options most affect Spring's behavior in production?

Q10 What mistakes do teams commonly make when adopting Spring?



ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 Spring is a tool or platform that

Mitigation

Spring is a tool or platform that automates or simplifies a specific concern in the Backend Technologies domain, reducing manual effort and operational complexity for engineering teams.

A6 Spring provides HA through leader election, replication, and observability

Spring provides HA through leader election, replication, or stateless horizontal scaling. Deploy at least two instances across availability zones and configure health checks for automatic failover.

A2 Spring has key building blocks that work

Primitives

Spring has key building blocks that work together: a configuration layer defines intent, a runtime layer enforces it, and an API or CLI layer allows operators to manage and observe the system.

A7 Spring exposes metrics (Prometheus endpoint or built-in dashboard)

Spring exposes metrics (Prometheus endpoint or built-in dashboard), structured logs, and optionally distributed traces. Define SLOs on the key metrics and alert before users are affected.

A3 Spring trades off simplicity against feature richness

Hardening

Spring trades off simplicity against feature richness differently than its alternatives. Choose Spring when its specific strengths performance, ecosystem, or ease of use align with your team's primary constraints.

A8 Performance bottlenecks typically stem from misconfigured connection

Performance bottlenecks typically stem from misconfigured connection pools, large payload sizes, insufficient worker threads, or missing caches. Profile with built-in diagnostics before optimizing.

A4 Install Spring via its package manager or

Gotchas

Install Spring via its package manager or binary release. Follow the quickstart to initialize a project, configure the minimal required settings, and run the default command to verify the setup works end-to-end.

A9 Run Spring with the principle of least

Assessment

Run Spring with the principle of least privilege, enable TLS for all communication, use a secrets manager for credentials, and keep the software patched to the latest stable release.

A5 Production-critical settings typically include concurrency and

Concurrency

Production-critical settings typically include concurrency limits, timeout values, retry policies, resource limits, and logging levels. Review the official production hardening guide before deploying.

A10 Common pitfalls include skipping the documentation and

Common pitfalls include skipping the documentation and learning the API by trial and error, using default configurations in production, lacking a runbook for operational issues, and not testing failover scenarios.